DO NOW · AO1 REVISION

Sound waves and waves for detection and exploration

Waves
Paper 2 · Separate Physics only
21 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 6 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**A sound wave passes into a different medium. Which stays the same: speed, frequency or wavelength?**	Frequency	☐
2	**State what happens to the speed of sound when it passes from air into a solid.**	It increases.	☐
3	**State what happens to the loudness of a sound passing from air into a solid, and why.**	It decreases: some sound is reflected or absorbed at the boundary.	☐
4	**Name the part of the human ear that detects sound waves.**	The eardrum	☐
5 ★	**State the range of human hearing.**	20 Hz to 20 kHz	☐
6 ★	**Name sound with a frequency above the range of human hearing.**	Ultrasound	☐
7	**What is echolocation?**	Using reflected sound (echoes) to locate an object.	☐
8	**Give two uses of ultrasound imaging.**	Medical scans (e.g. of an unborn baby) and finding flaws in materials	☐
9	**What are seismic waves?**	Waves produced by earthquakes.	☐
10	**Name the two types of seismic waves.**	P-waves and S-waves	☐
11 ★	**What type of wave are P-waves?**	Longitudinal	☐
12	**Name the states of matter that P-waves travel through.**	Solids and liquids	☐

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	QUESTION	ANSWER	✓
13 ★	**What type of wave are S-waves?**	Transverse	☐
14	**Name the state of matter that S-waves travel through.**	Solids only	☐
15	**Describe how a sound wave travels through a solid.**	The particles of the solid vibrate and pass the vibrations on.	☐
16	**Explain why human hearing is limited to a range of frequencies.**	Sound makes the eardrum vibrate only over a limited range of frequencies.	☐
17	**State what happens to ultrasound at a boundary between two media.**	It is partially reflected.	☐
18 ★	**Describe how ultrasound is used to find the distance to a boundary.**	Time the reflection: distance = speed × time ÷ 2.	☐
19	**State what echo sounding is used for.**	Detecting objects in deep water and measuring water depth.	☐
20	**State what seismic waves have provided evidence for.**	The structure and size of the Earth's core.	☐
21 ★	**Explain how S-waves provide evidence that part of the Earth's core is liquid.**	S-waves cannot travel through liquid and are not detected on the far side of the Earth.	☐

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